

STATS 200A Discussion 1

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29 September 2026

$A = \{6\}, \mathbb{P}(A) = \frac{1}{6}$
 For a dice, $\Omega = \{1, 2, 3, 4, 5, 6\}$
 For a coin, $\Omega = \{H, T\}$
 For 2 dice,

$$\Omega = \left\{ \begin{pmatrix} (1,1) & (1,2) & \dots & (1,6) \\ & & & \vdots \\ & & & (6,6) \end{pmatrix} \right\}$$

$$\mathcal{F} : \begin{cases} \emptyset, \Omega \in \mathcal{F} \\ A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F} \\ A_1, A_2, A_3, \dots \in \mathcal{F} \Rightarrow \bigcup_{n=1}^{\infty} A_n \in \mathcal{F} \end{cases}$$

$$\mathcal{F} = \{\emptyset, \Omega\}$$

$$\mathcal{P} = 2^{\Omega}$$

Example 1.1

$$\Omega = \{1, 2, 3, 4\}$$

$$\mathcal{F} = \{\emptyset, \Omega, \{1, 2\}, \{3, 4\}\}$$

Solution. $\emptyset, \Omega \in \mathcal{F}$

$$A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F}$$

$$\{1, 2\} \cup \{3, 4\} = \Omega$$

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$$A \subseteq \Omega$$

$$\sigma(A) = \mathcal{F} : A \in \mathcal{F} \text{ smallest: } \exists \mathcal{F}' : A \in \mathcal{F}' \text{ then } \mathcal{F} \subseteq \mathcal{F}'$$

$$\sigma(A) = \bigcap_{A \in \mathcal{F}} \mathcal{F}$$

Example 1.2

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

$$A = \{1, 2, 3\}$$

$$B = \{1, 3, 5\}$$

Solution. $\sigma(A) = \{\emptyset, \Omega, \{1, 2, 3\}, \{4, 5, 6\}\}$

$$\sigma(A, B) = \{\emptyset, \Omega, \{1, 2, 3\}, \{1, 3, 5\}, \{4, 5, 6\}, \{2, 4, 6\}, \{1, 3\}, \{5\}, \{2\}, \{4, 6\}, \{1, 2, 3, 4, 6\}, \{1, 3, 4, 5, 6\}$$

$$\{2, 4, 5, 6\}, \{1, 2, 3, 5\}, \{1, 3, 4, 6\}, \{2, 5\}\}$$

$$A, B \in \mathcal{F} \quad A \cap B = (A^c \cup B^c)^c$$

$$A/B = A \cap B^c \in \mathcal{F}$$

Take the smallest partitions: $\{1, 3\}, \{5\}, \{2\}, \{4, 6\}$ and take the power set to construct the Borel σ -algebra

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Example 1.3

$$\begin{aligned}\Omega &= \{(1, 2, 2, 6, 1, 3, \dots)\} \\ &= \{(a_1, a_2, a_3, \dots) \mid a_i \in \{H, T\}\} \\ &= \{H, T\}^{\mathbb{N}}\end{aligned}$$

Defined H_n as $\{n^{\text{th}} \text{ toss is Head}\}$
 $H_3 = \{(a_1, a_2, H, a_4, \dots) \mid a_i \in \{H, T\}\}$
 $A = \{\text{Heads occurs at least once}\}$

Solution. $A = \bigcup_{n=1}^{\infty} H_n$

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Example 1.4

$B = \{\text{Heads occurs only before or on the 10th toss}\}$

Solution. $B = (\bigcup_{n=11}^{\infty} H_n)^c$

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Example 1.5

$C = \{\text{Heads occurs only finitely many times}\}$

Solution.

$$C = \left(\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} H_k \right)^c$$

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